

Jiayao (Claris) Gu

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LinkedIn • Ph.D. candidate @ Mila and @ McGill • Homepage @ <https://jiayaogu.github.io>

Biography

I am a Ph.D. candidate at **Mila – Québec AI Institute** and **McGill University**, specializing in **Generative AI**, **Large Language Models** and their applications on **Science**. Representative contributions include (1) a **first-author paper** introducing a dual-transformer-encoder framework accepted at ISMB (16% acceptance rate); (2) a **co-first-authored paper** on model-based optimization with Yoshua Bengio, awarded the **Survey Certificate** at TMLR 2026; and (3) a co-authored **ICML 2026 spotlight paper** (top 2.2%).

My research excellence is recognized through the **DNA2RNA Scholarship** (\$80,000), the **FRQ-NT Scholarship** (\$26,666) for genome-based therapies, the **Tinker Research Grant** (\$5,000), and **Cohere Labs' Catalyst Grants**. Previously, I received my B.Sc. in **Computer Science and Biology** from McGill University, graduating on the **Dean's Honour List** (top 10%). Beyond academia, I have conducted LLM research at **Noah's Ark Lab**, and served as an intern supervisor at **Tiptree Systems**. I also received the **Rising Star Award** at the McMedHacks Hackathon for a 3D medical image segmentation challenge.

Education

Mila - Québec AI Institute / McGill University

Sep 2024 – Present

Ph.D. Computer Science (GPA: 3.86/4.0)

Montréal, QC, Canada

Supervisor: Prof. Yue Li ([homepage](#))

Sponsored by the prestigious **DNA2RNA Scholar Award** (\$80,000) for exceptional graduates.

McGill University

Sep 2022 – Sep 2024

MSc. Computer Science

Montréal, QC, Canada

Awarded **FRQNT Research Scholarship** (\$26,666) recognizing top Québec graduate students.

Received **Graduate Excellence Award** (\$33,000) for high-performing graduate students.

McGill University

Sep 2018 – Apr 2022

BSc. Computer Science and Biology

Montréal, QC, Canada

Recognized on the **Dean's Honour List** (Top 10% of the faculty).

Experience

Noah Ark Lab

May 2026 - Present

Research Intern

Montréal, Québec, Canada

- Built large-scale benchmarks for test-time skill adaptation. Helped build a 512-task benchmark by injecting hidden task rules into six existing large-scale datasets, enabling systematic evaluation of whether agents can learn from prior experience and adapt to task-skill gaps at test time.
- Identified limitations of existing memory/skill adaptation methods. Through 20+ literature review and extensive reproduction and failure analysis of 9+ top ML conference papers, identified limitations in how existing methods store, retrieve, update, and transfer past experience.
- Contributed to the design and validation of our memory/skill adaptation method. Based on the baseline analysis, contributed to the core design of our method for jointly improving agentic skill storage, effective retrieval, and transfer adaptation.

Tiptree Systems

May 2026 - Jan 2027

Intern supervisor

Montréal, Québec, Canada

- Offer valuable support with my expertise in machine learning to build agentic system that verify claims of paper reviewers.

Mila - Québec AI Institute | McGill University

Jan 2024 - Present

Research Assistant

Montréal, Québec, Canada

- Leading research in biological molecule synthesis and function prediction [2].
- Leading scientific agent development.

McGill University

Aug 2021 - Aug 2024

Graduate Researcher

Montréal, Québec, Canada

- Developed and optimized an 3D CT image segmentation workflow using state-of-the-art CNN framework, improving prediction accuracy by 25%.
- Published findings in high-impact peer-reviewed journals [7] [8].

Hackathon Team Leader

Montréal, Québec, Canada

- McMedHacks (2021): Developed an RCNN-based solution for 3D CT image segmentation, earning the Rising Star Award.
- McGill CodeJam (2022): Created a machine-learning backend for ad impression prediction, winning Broadsign challenge prize.

Preprints

* denotes co-first author.

- [1] **Jiayao Gu**, Yue Li. A generative-AI framework for target-Specific MicroRNAs towards RNAi-based drug design (2026) bioRxiv 2026.05.07.723585; doi:<https://www.biorxiv.org/content/10.64898/2026.05.07.723585v1> (Submitted to PLOS Computational Biology)

Publications

* denotes co-first author.

- [2] **Jiayao Gu**, Can (Sam) Chen, Yue Li. MiRformer: A Dual-Transformer-Encoder Framework for Predicting MicroRNA-MRNA Interactions from Paired Sequences (2026) bioRxiv 2025.11.21.689769; doi: <https://doi.org/10.1101/2025.11.21.689769> (**Accepted to ISMB, acceptance rate=65/408=16%**)
- [3] Zipeng Sun, Can Chen, Ye Yuan, Haolun Wu, **Jiayao Gu**, Christopher Pal, Xue Liu. Training Diffusion Language Models for Black-Box Optimization (2026) arXiv, https://openreview.net/forum?id=Z7wIOsor6i&referrer=%5Bthe%20profile%20of%20Xue%20Liu%5D%28%2Fprofile%3Fid%3D~Xue_Liu1%29 (Accepted to **ICML2026 Spotlight Top 2.2%**)
- [4] Minsu Kim*, **Jiayao Gu***, Ye Yuan, Taeyoung Yun, Yoshua Bengio, Can (Sam) Chen. Offline Model-Based Optimization: Comprehensive Review (2025). TMLR: <https://openreview.net/pdf?id=QcSZWo1TL1> (**Awarded Survey Certificate for papers that provide an exceptionally thorough survey of the topic**)
- [5] **Jiayao Gu**, Yihong Li. The Role of Data Selection in Language Model Training (2025). ICML 2025 Workshop: <https://arxiv.org/abs/2501.03826>
- [6] **Jiayao G.***, Kexin C.*, Peidong L., Yue, Y., Lynn A., Yang L., Tianyu S. ClawBot-Matching: Bidirectional, Explainable, and Learnable Collaboration Matching in Mixed Human-Agent Networks (2026). RLEval: Methods and Reinforcement Learning Environments for Evaluating AI Agents. <https://openreview.net/forum?id=6NTHOH5Loo>
- [7] Jamie Magrill, Dan Moldoveanu, **Jiayao Gu**, Matieu Lajoie, Ian R. Watson. Mapping the Single Cell Spatial Immune Landscapes of the Melanoma Microenvironment (2023). Clinical & Experimental Metastasis: <https://link.springer.com/article/10.1007/s10585-023-10252-4>
- [8] Magrill J*, **Gu Jiayao***, Allard-Puscas S., Ramsay L., Lajoie M., Karimi E., Maritan S.M., Dankner M., Rezanejad M, Zuo D.M., Wei Y.H., Petrecca K., Guiot M.C., Routy B., Siegel P.M, Quail D., Walsh L., Liviu C., Mihalciou D., Stagg J., and Watson IR. Spatially Mapping the Single-Cell Immune Landscapes of Melanoma Brain Metastases Using Imaging Mass Cytometry (2024). Journal of Clinical Oncology: https://ascopubs.org/doi/10.1200/JCO.2024.42.16_suppl.e21517

Awards and Scholarships

D2R Doctoral Scholar Award

2026

Amount: \$80,000

Scope: National

Description: Doctoral Award for exceptional graduate researchers in Canada.

ISMB 2026 Conference Fellowship

2026

Amount: \$750

Scope: Global

Description: Fellowship awarded to students to present their papers at ISMB conference.

Laura Bassi Scholarship	2026
Amount: \$500	
Scope: Global	
Description: Award to support exceptional work undertaken by all graduates.	
DNA2RNA (D2R) Scholar Award	2025
Amount: \$27,000	
Scope: Provincial	
Description: Highest recognition for D2R trainees; awarded to outstanding and active researchers in genomic-based therapeutics.	
FRQNT Research Scholarships	2023
Amount: \$26,666	
Scope: Provincial (Québec)	
Description: Research Scholarships sponsored by Fonds de recherche du Québec - Nature et Technologies (FRQNT) by the Québec government, recognizing excellent graduate students in Québec.	
Grad Excellence Award	2022
Amount: \$33,000	
Scope: Institutional (McGill University)	
Description: Awarded by McGill University to top-performing graduate students.	
McGill CodeJam Challenge Winner	2022
Amount: \$600	
Scope: Institutional (McGill University)	
Description: 1st Place Winner of the Broadsign machine learning challenge for ad impression prediction.	
Rising Star Award in 2021 McMedHacks Medical Image Hackathon	2021
Amount: \$100	
Scope: Institutional (McGill University)	
Description: McMedHacks 2021 Medical Image Hackathon for 3D CT image segmentation innovation	
Dean's Honour List (Top 10%)	2019
Scope: Institutional (McGill University)	
Description: Recognized as the top 10% in the faculty based on CGPA.	

Computer Science Skills

Programming: Python, Java, C, C++, R, bash scripting, HTML
Machine Learning Frameworks: PyTorch, TensorFlow, JAX, scikit-learn
Cloud Computing: GCP, AWS
Tools: Skilled in using Git/GitHub, MATLAB